

Guidelines for Peptide-level scoring

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<SpectrumIdentificationProtocol analysisSoftware_ref="ID_software" id="SearchProtocol_1">
  <SearchType>
    <cvParam accession="MS:1001083" cvRef="PSI-MS" name="ms-ms search"/>
  </SearchType>
  <AdditionalSearchParams>
    <cvParam accession="MS:1001211" cvRef="PSI-MS" name="parent mass type mono"/>
    <cvParam accession="MS:1001256" cvRef="PSI-MS" name="fragment mass type mono"/>
    <cvParam accession="MS:1002490" cvRef="PSI-MS" name="peptide-level scoring"/>
    <cvParam accession="MS:1002496" cvRef="PSI-MS" name="group PSMs by sequence"/>
  </AdditionalSearchParams>
  <ModificationParams>
    <SearchModification residues="C" massDelta="57.021465" fixedMod="true">
      <cvParam accession="UNIMOD:4" cvRef="UNIMOD" name="Carbamidomethyl"/>
    </SearchModification>
    <SearchModification residues="M" massDelta="15.994915" fixedMod="false">
      <cvParam accession="UNIMOD:35" cvRef="UNIMOD" name="Oxidation"/>
    </SearchModification>
  </ModificationParams>
  <Enzymes independent="false">
    <Enzyme missedCleavages="1" semiSpecific="false" cTermGain="OH" nTermGain="H" id="Enz1">
      <EnzymeName>
        <cvParam accession="MS:1001251" cvRef="PSI-MS" name="Trypsin"/>
      </EnzymeName>
    </Enzyme>
  </Enzymes>
  <FragmentTolerance>
    <cvParam accession="MS:1001412" cvRef="PSI-MS" unitCvRef="UO" unitName="dalton" unitAccession="UO:0000221" value="0.8" name="search tolerance plus value"/>
    <cvParam accession="MS:1001413" cvRef="PSI-MS" unitCvRef="UO" unitName="dalton" unitAccession="UO:0000221" value="0.8" name="search tolerance minus value"/>
  </FragmentTolerance>
  <ParentTolerance>
    <cvParam accession="MS:1001412" cvRef="PSI-MS" unitCvRef="UO" unitName="dalton" unitAccession="UO:0000221" value="1.5" name="search tolerance plus value"/>
    <cvParam accession="MS:1001413" cvRef="PSI-MS" unitCvRef="UO" unitName="dalton" unitAccession="UO:0000221" value="1.5" name="search tolerance minus value"/>
  </ParentTolerance>
  <Threshold>
    <cvParam accession="MS:1001868" cvRef="PSI-MS" name="distinct peptide-level q-value" value="0.01"/>
    <cvParam accession="MS:1002354" cvRef="PSI-MS" name="PSM-level q-value" value="0.01"/>
  </Threshold>
</SpectrumIdentificationProtocol>
```

A

B

C

```
<SpectrumIdentificationResult spectralData_ref="SID_1" spectrumID="index=145" id="SIR_5">
  <SpectrumIdentificationItem passThreshold="false" rank="1" peptide_ref="SSHAPVPHGVRLWK" calculatedMassToCharge="523.284" experimentalMassToCharge="523.194" chargeState="3" id="SIR_5_1">
    <PeptideEvidenceRef peptideEvidence_ref="PE5_2_9"/>
    <cvParam accession="MS:1001328" cvRef="PSI-MS" value="3.05370337630321" name="OMSSA:evale"/>
    <cvParam accession="MS:1001329" cvRef="PSI-MS" value="6.82544339808495E-4" name="OMSSA:pvalue"/>
    <cvParam accession="MS:1002520" cvRef="PSI-MS" value="SSHAPVPHGVRLWK" name="peptide group ID"/>
    <cvParam accession="MS:1001868" cvRef="PSI-MS" value="distinct peptide-level q-value" name="6.82544339808495E-4"/>
    <cvParam accession="MS:1002500" cvRef="PSI-MS" value="peptide passes threshold" name="true"/>
  </SpectrumIdentificationItem>
  <cvParam accession="MS:1000796" cvRef="PSI-MS" value="55.6021.6024.3.dta" name="spectrum title"/>
</SpectrumIdentificationResult>
<SpectrumIdentificationResult spectralData_ref="SID_1" spectrumID="index=121" id="SIR_6">
  <SpectrumIdentificationItem passThreshold="false" rank="1" peptide_ref="SSHAPVPHGVRLWK" calculatedMassToCharge="523.284" experimentalMassToCharge="523.194" chargeState="3" id="SIR_6_1">
    <PeptideEvidenceRef peptideEvidence_ref="PE5_2_9"/>
    <cvParam accession="MS:1001328" cvRef="PSI-MS" value="4.05370337630321" name="OMSSA:evale"/>
    <cvParam accession="MS:1001329" cvRef="PSI-MS" value="7.82544339808495E-4" name="OMSSA:pvalue"/>
    <cvParam accession="MS:1002520" cvRef="PSI-MS" value="SSHAPVPHGVRLWK" name="peptide group ID"/>
    <cvParam accession="MS:1001868" cvRef="PSI-MS" value="distinct peptide-level q-value" name="6.82544339808495E-4"/>
    <cvParam accession="MS:1002500" cvRef="PSI-MS" value="peptide passes threshold" name="true"/>
  </SpectrumIdentificationItem>
  <cvParam accession="MS:1000796" cvRef="PSI-MS" value="55.6021.6025.3.dta" name="spectrum title"/>
</SpectrumIdentificationResult>
```

D

E

F

D

E

F

Feature	Explanation
A	cvParam indicating that peptide-level scoring has been done and that feature B MAY be present and features D, E and F MUST be present. A cvParam MUST also be present indicating the type of grouping of PSMs to peptides done.
B	The threshold used to determine whether each distinct peptides group has been confidently identified – used to set the value of feature F
C	As for regular mzIdentML files, a threshold can be applied at the level of PSMs – which is used to set the passThreshold attribute on <SpectrumIdentificationItem>
D	If feature A is present, exactly one cvParam “peptide group ID” MUST be present in which the value slot contains a unique identifier (string) that MUST be given to all <SpectrumIdentificationItem>elements within the same distinct peptide group. There is no expectation that meaningful information SHOULD be conveyed by the value slot, but implementers MAY choose to use the peptide sequence or peptide sequence and a modification string (depending on the grouping mechanism) as the value.
E	A cvParam containing the peptide-level score used for ordering <i>distinct peptide</i> entities, which MUST be given to all <SpectrumIdentificationItem>elements within the same distinct peptide group with the same value.
F	If feature A is present, every <SpectrumIdentificationItem>element MUST contain the “peptide passes threshold” cvParam with a Boolean value. All <SpectrumIdentificationItem>elements within the same distinct peptide group MUST have the same value.